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"""
EVENTVERSE - Event and Festival
Announcement System
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"""

import json
from datetime import datetime

FILE_NAME = "events.json"

# ----- File handling -----
def load_events():
    try:
        with open(FILE_NAME, "r") as f:
            return json.load(f)
    except (FileNotFoundError,
            json.JSONDecodeError):
        return []

def save_events(events):
    with open(FILE_NAME, "w") as f:
        json.dump(events, f, indent=4)

# ----- Helper functions -----

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def next_id(events):
    if not events:
        return 101
    return max(e["event_id"] for e in
events) + 1

def get_valid_date(prompt):
    while True:
        value = input(prompt).strip()
        try:
            datetime.strptime(value, "%Y-
%m-%d")
            return value
        except ValueError:
            print("Invalid date. Please use
YYYY-MM-DD (e.g. 2026-11-08).")

def find_event(events, event_id):
    for e in events:
        if e["event_id"] == event_id:
            return e
    return None

def ask_event_id():
    try:
        return int(input("Enter event ID:
"))
    except ValueError:
        print("Event ID must be a number.")
        return None
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def show_event(e):
    print("-" * 55)
    print(f"[{e['event_id']}] {e['title']}
({e['category']})")
    print(f"  Description :
{e['description']}")
    print(f"  Date       : {e['date']}")
    print(f"  Time       : {e['time']}")
    print(f"  Venue       : {e['venue']},
{e['city']}")
    print(f"  Organizer  :
{e['organizer_name']}")
    print(f"  Contact    :
{e['organizer_contact']}")
    print(f"  Participants:
{len(e['participants'])}")

# ----- Main features -----
def add_event(events):
    print("\n--- Add New Event ---")
    event = {
        "event_id": next_id(events),
        "title": input("Event/Festival
name: ").strip(),
        "category": input("Category
(Festival/Cultural/Sports/Technical):
").strip(),
        "description": input("Short
description: ").strip(),
        "date": get_valid_date("Date (YYYY-
MM-DD): "),
        "time": input("Timing (e.g. 06:00

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PM - 10:00 PM): ").strip(),
    "venue": input("Venue: ").strip(),
    "city": input("City: ").strip(),
    "organizer_name": input("Organizer
name: ").strip(),
    "organizer_contact":
input("Organizer contact (phone/email):
").strip(),
    "participants": [],
}
events.append(event)
save_events(events)
print(f"Event added successfully! Event
ID = {event['event_id']}")

def view_events(events):
    print("\n--- Upcoming Events ---")
    if not events:
        print("No events announced yet.")
        return
    for e in sorted(events, key=lambda x:
x["date"]):
        show_event(e)
    print("-" * 55)

def search_events(events):
    print("\n--- Search Events ---")
    key = input("Enter keyword (name / city
/ venue / category / date):
").strip().lower()
    found = [
        e for e in events

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        if key in e["title"].lower() or key
in e["city"].lower()
        or key in e["venue"].lower() or key
in e["category"].lower()
        or key in e["date"]
    ]
    if not found:
        print("No matching events found.")
        return
    print(f"{len(found)} event(s) found:")
    for e in found:
        show_event(e)

def register_participant(events):
    print("\n--- Register Participant ---")
    event_id = ask_event_id()
    if event_id is None:
        return
    e = find_event(events, event_id)
    if e is None:
        print("Event not found.")
        return
    name = input("Participant name:
").strip()
    contact = input("Participant contact:
").strip()
    e["participants"].append({"name": name,
"contact": contact})
    save_events(events)
    print(f"{name} registered for
'{e['title']}'")

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def view_participants(events):
    print("\n--- Participant List ---")
    event_id = ask_event_id()
    if event_id is None:
        return
    e = find_event(events, event_id)
    if e is None:
        print("Event not found.")
        return
    print(f"Event: {e['title']} | Total
participants: {len(e['participants'])}")
    if not e["participants"]:
        print("No participants registered
yet.")
        return
    for i, p in
enumerate(e["participants"], start=1):
        print(f"  {i}. {p['name']} -
{p['contact']}")

def update_event(events):
    print("\n--- Update Event ---")
    event_id = ask_event_id()
    if event_id is None:
        return
    e = find_event(events, event_id)
    if e is None:
        print("Event not found.")
        return
    print("Press Enter to keep the old
value.")
    e["title"] = input(f"Name
[{e['title']}] : ").strip() or e["title"]

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        new_date = input(f>Date [{e['date']}]
(YYYY-MM-DD): ").strip()
        if new_date:
            try:
                datetime.strptime(new_date,
"%Y-%m-%d")
                e["date"] = new_date
            except ValueError:
                print("Invalid date, old date
kept.")
            e["time"] = input(f>Time [{e['time']}]
").strip() or e["time"]
            e["venue"] = input(f>Venue
[{e['venue']}]
").strip() or e["venue"]
            e["organizer_contact"] = (
                input(f>Organizer contact
[{e['organizer_contact']}]
").strip()
                or e["organizer_contact"]
            )
            save_events(events)
            print("Event updated successfully!")

def delete_event(events):
    print("\n--- Cancel / Delete Event ---
")
    event_id = ask_event_id()
    if event_id is None:
        return
    e = find_event(events, event_id)
    if e is None:
        print("Event not found.")
        return
    confirm = input(f>Delete

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    '{e['title']}'? (y/n): ").strip().lower()
    if confirm == "y":
        events.remove(e)
        save_events(events)
        print("Event deleted.")
    else:
        print("Cancelled.")

# ----- Program start -----
def main():
    events = load_events()
    while True:
        print("\n" + "=" * 40)
        print("          EVENTVERSE")
        print(" Event & Festival
Announcement System")
        print("=" * 40)
        print("1. Add new event")
        print("2. View all upcoming
events")
        print("3. Search events")
        print("4. Register participant")
        print("5. View participants of an
event")
        print("6. Update event")
        print("7. Cancel / delete event")
        print("8. Exit")
        choice = input("Enter your choice
(1-8): ").strip()

        if choice == "1":
            add_event(events)
        elif choice == "2":

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        view_events(events)
    elif choice == "3":
        search_events(events)
    elif choice == "4":
        register_participant(events)
    elif choice == "5":
        view_participants(events)
    elif choice == "6":
        update_event(events)
    elif choice == "7":
        delete_event(events)
    elif choice == "8":
        print("Thank you for using
Eventverse. Goodbye!")
        break
    else:
        print("Invalid choice. Please
enter a number from 1 to 8.")

if __name__ == "__main__":
    main()
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